

Aditya Mukhopadhyay

AI Engineer – Badho Technologies
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EDUCATION

B.Tech, Information Technology
IGEC Sagar (RGPV), Madhya Pradesh

2021 – 2025 · CGPA 7.5

Class XII (PCM)

SJC S.S. School, Ranjhi, Jabalpur
2021 · 93.4%

LINKS

Portfolio

adityamukhopadhyay.vercel.app

LinkedIn

adityamukhopadhyay2003

GitHub

adityamukhopadhyay

SKILLS

Core

Python · TypeScript · SQL · FastAPI · PostgreSQL · Supabase · Docker · AWS (EC2, Lambda, S3) · Railway · Vercel

AI / LLM

Claude Agent SDK · Model Context Protocol (MCP) · LangGraph · Claude / GPT / Gemini APIs · FAISS · embeddings & RAG · prompt caching & cost optimization · agent guardrails & evals · n8n

PROJECTS

Aditya MCP – Agent-Ops Infrastructure

2026

Read-only **MCP server** mining AI coding-session transcripts, git history and handoff docs across every project; powers **scheduled autonomous routines** – daily work reports and cross-session stale-doc nudges. Agents auditing agents.

Serverless Multimodal Call QA
2025

AWS Lambda + Gemini 2.5 Pro pipeline auditing recorded sales calls, audio fed straight to the model – **100% call coverage**, structured JSON verdicts to Postgres/Metabase, replacing manual QA.

SUMMARY

AI Engineer building production multi-agent systems and LLM infrastructure: autonomous agent fleets (Claude Agent SDK), Model Context Protocol (MCP) servers, conversational AI over WhatsApp, and semantic retrieval (FAISS/RAG) – with deterministic guardrails, honest evaluation, and cost engineering that holds up in production.

WORK EXPERIENCE

AI Engineer | Badho Technologies Pvt. Ltd.

Jun 2025 – Present

• Autonomous Delivery Fleet – Multi-Agent Operations (Claude Agent SDK, MCP)

- Built a production fleet where **one AI agent supervises each rider's full delivery day**: 20+ concurrent Claude Agent SDK sessions read live PostgreSQL via an in-process **MCP** toolset, coach riders over WhatsApp in Hinglish, file Freshdesk tickets, and escalate only true anomalies.
- Enforced safety structurally – recipient guards, invented-number blocking, watchdogs, healthchecks – backed by **23 offline test suites (~800 assertions)**; designed a **rule/LLM hybrid** that runs routine nudges as deterministic templates re-verified at send time, reserving the model for judgement and replies, so cost tracks difficulty rather than traffic.

• Conversational Commerce – WhatsApp AI Sales Agent + 100-Tool Buyer MCP

- Authored an **MCP server exposing the buyer app as 100 tools** (Hasura GraphQL + REST, phone-OTP OAuth, per-buyer JWT) with exact app-parity cart, coupon and deeplink engines – consumed by production agents, Claude Desktop and Claude Code.
- Shipped the live WhatsApp sales bot on top: tool-using worker + tone-controlled reply agent, with deterministic honesty gates – it can never claim an unplaced order, and ticket raises are hard-gated against existing tickets.
- Rebuilt attribution after measuring that **75% of credited orders predated first contact**, replacing the vanity metric with exact cart-ID causal matching; every tile reconciles **exactly** against the production replica (**INR 5.0L across 475 orders**) at **\$152.50 total LLM spend**.

• Order-Lifecycle WhatsApp Automation (event-driven, incident-hardened)

- Event-time-guarded notification engine – **10 order stages, ~500 messages/day, zero errors over 8 days**; built the guard architecture post-incident so bulk-touched rows can never trigger sends.

• Delhivery Courier Ops Autopilot – Claims & Tickets (MCP + Dashboard)

- Autonomous recovery loop over our courier partner Delhivery's ops API: auto-files **weight-discrepancy claims** and **NDR/lost/damaged tickets**, triaged by Claude Haiku with **vision on exception images** – **698 claims + 370 tickets filed in 24h, 2,143 closed** to date; a **session-liveness kill switch** halts all loops and AI spend when the upstream session dies.

• Search, Matching & Catalog Intelligence

- Sub-100ms semantic search (**FAISS** + Ollama embeddings + Postgres phonetics); sister-product clustering across **85,000+ SKUs** (HDBSCAN/K-Means); **170,000+ external SKUs** matched via a cost-tiered funnel (Typesense → trigram → batched LLM).

• Rewards & Growth Platforms (live money paths)

- Built and operate real-money systems end-to-end: a referral platform – **Next.js buyer webview** (app-injected auth, one link for every buyer), **Supabase credit service**, poller releasing rewards only on **delivered** orders, **AiSensy WhatsApp** notifications – **361/361 bonuses credited, zero misses**; plus QR scratch cashback with **DB-level double-credit prevention**.

AI Developer (Freelance) | Strix Digital

Jul 2023 – May 2025

- Built AI web-apps (custom **Cold Email Generator** on **Groq**), automation tooling and scrapers; shipped an iOS app for a Houston client.

Data Science Intern | Acmegrade (in collab. with IIT Bombay)

Apr 2022 – Jun 2022

- Built ML models for **movie recommendation**, **sales forecasting** and **CNN sound classification**; EDA and feature engineering on multi-modal data.